

Project Module



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Algae extracts as bio-effectors for improved growth, mineral nutrition and stress resilience of Durum wheat

Plants effectors-a novel product category for agriculture

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DECLARATION

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List of Abbreviations

EU: European Union

FAO: Food and Agriculture Organization

N: Nitrogen

P: Phosphorus

K: Potassium

Mg: Magnesium

Mn: Manganese

Ca: Calcium

Cu: Copper

S: Sulfur

Zn: Zinc

B: Boron

VDLUFA: Verband Deutscher Landwirtschaftlicher Untersuchungs- und Forschungsanstalten

SAS: Statistical Analysis System

LSD: Least Significant Difference

AAS: Atomic Absorption Spectroscopy

SPAD: chlorophyll meter “soil plant analysis development”

SWP4: Seaweed produced by wet milling process

SPAP4: Freshwater algae produced by wet milling process

SPA+BEM: Freshwater algae produced by fermentation

SPB24P5: Freshwater algae produced by wet milling

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0. Abstract

The entire agricultural sector is putting forward the sustainable agricultural practices and innovative approaches to enhance crop productivity without compromising the resources for the future. This study investigated the efficacy of seaweed algae-based bio-effectors in improving growth, nutrient use efficiency, and stress resilience in *Triticum durum* (Durum wheat), a crop highly sensitive to abiotic stressors such as nutrient deficiencies and drought.

A first filter paper screening tests were first conducted to identify promising bio-effector products and their effective concentrations to improve the germination, root and shoot growth of wheat seedlings. Following this preliminary evaluation, a controlled pot experiment was carried out in a greenhouse at the University of Hohenheim. The study involved eight levels of bio-effector treatments inclusive the untreated control variant applied to a single spring durum wheat variety (Beladur), with five replications, resulting in a total of 40 experimental units. The bio-effector treatments included novel seaweed-derived bio-effector products developed by the non-profit organization Open Climate Solutions (SPAP4, SPB24P5, SPA+BEM and SWP4) as well as two commercially available reference products (AMLLK and Ukelp). This stepwise approach was chosen to ensure a rigorous assessment of the bio-effectors' potential under controlled conditions. During the growth phase of the durum wheat plants, a deliberate drought stress phase with subsequent recovery phase under adequately watered conditions was implemented to expose the plants to abiotic stress. By simulating drought conditions, the study provided a controlled environment to assess the efficacy of the BEs in mitigating the adverse effects of water deficiency.

For experimental evaluations, plant height, water consumption, chlorophyll index, biomass, and mineral nutrient concentrations in plant tissues were measured and analysed and the data were analyzed using the statistical software programs SAS and R-Software. The results revealed that SPAP4 increased root biomass significantly, achieving 3.79 g compared to 2.07 g dry matter per pot in the control ($P \leq 0.05$), while SPAP4 and AMLLK maintained the highest chlorophyll indices under drought stress, with SPAD values of 46.05 and 44.92, respectively, compared to 34.65 in the control. Water consumption during the drought phase was reduced in bio-effector treatments, with SPA+BEM, AMLLK, and Ukelp recording 762.2 g, 761.0 g, and 760.0 g, respectively, compared to 924.8 g per pot in the control ($P \leq 0.05$), as recorded during the time frame of 70 days until 93 days after sowing. Despite the enhanced root biomass, drought resilience, and chlorophyll retention under water-limited conditions, plant height, plant biomass production, mineral nutrient

concentrations in plant tissues, and the accumulation of mineral nutrients in the plant biomass showed no statistically significant differences.

In addition to the pot experiment, a field trial with winter wheat was conducted in collaboration with Farm Gut Wilhelmsthal in Saxony-Anhalt. The study tested the commercial seaweed products Ukelp and TopHealth during the 2024 growing season, a period marked by record-high temperatures and severe drought - the most extreme conditions since weather data recording began in 1881.

The field trial results revealed a promising, though statistically insignificant, 3% increase in grain yield, with Ukelp and TopHealth producing 6.67 t ha⁻¹ compared to 6.49 t ha⁻¹ in the control. TopHealth demonstrated significantly higher above-ground biomass, with straw and ear yields of 6.61 t ha⁻¹ and 9.71 t ha⁻¹, respectively, compared to Ukelp (6.37 t ha⁻¹ and 9.53 t ha⁻¹) and the control (6.20 t ha⁻¹ and 9.44 t ha⁻¹).

These findings underscore the role of bio-effectors in modulating plant responses, especially under abiotic stress, and their potential to enhance sustainable crop management practices. They align with the European Union's Farm-to-Fork strategy, promoting the reduction of synthetic agrochemical inputs while supporting sustainable intensification. The study highlights the need for further research to refine bio-effector application strategies, including optimal dosages, product combinations, and adapted nutrient and crop management, to maximize benefits across diverse cropping systems.

1. Introduction

1.1. Background

Globally, the world is experiencing three to five times increase in the production of staple cereals like wheat, maize and rice over the last 60 years, but there are also growing concerns regarding the sustainability and environmental effect of that intensive production (Soto-Gómez and Pérez-Rodríguez, 2022). Recently a decline in the growth rates of major crop productivity was detected by Food and Agriculture Organization (FAO) in both regions of fertile soil and less fertile, even though pesticides and mineral fertilisers are intensively used, due to nutrient deficiencies, and abiotic stress (Weinmann, 2019). According to Statista 2024, above 195 million tons of N fertilizer are annually applied in crop growing, but inadequate use of nitrogen fertilizers is an alarming threat to soil health and biodiversity (Statista 2024), and only 50 percent of the applied nitrogen is used effectively (Cassman et al., 2002). Faced with these challenges, the European Union, through its Farm to Fork Policy, aims to reduce agrochemical use by targeting a 50 % reduction in pesticide use, a 20 % reduction in fertilizer use, and at least a 50 % decrease in nutrient losses, all while ensuring that there is no loss of soil fertility. Eventually, bio-effectors, comprising beneficial microorganisms and active natural compounds, are one of the solutions for soil fertility, sustainability and climate change due to their ability to enhance nutrient uptake and improve stress tolerance in various crops (Billen et al., 2024).

Bio-effectors like seaweed extracts contain bioactive metabolites that positively affect plant metabolism and lead to the enhancement of plant growth and resilience under limited resources of mineral nutrients and water (Zhang et al., 2024; Bhattacharya et al., 2015; Vaghela et al., 2023).

Seaweed extracts achieve growth-promoting effects and enhance resilience by mimicking the action of phytohormones such as auxins, cytokinin, abscisic acid, and ethylene (Khan et al., 2009). While the phytohormone levels in seaweed extracts might be insufficient to directly induce significant effects at recommended application rates, their active components comprising polysaccharides, proteins, polyunsaturated fatty acids, pigments, and polyphenols may modulate innate pathways in plants to stimulate the biosynthesis of these hormones (Chojnacka et al., 2012; Wally et al., 2013). Auxins, for instance, enhance shoot vigor and root extension, while abscisic acid and ethylene improve resilience to abiotic stresses such as drought, temperature fluctuations, and humidity extremes. Cytokinin contributes by promoting cell division and multiplication, further supporting plant growth and development (Khan et al., 2009).

Durum wheat is one the crop plants which are particularly sensitive to limited resources of nutrients and water (Zhang et al.,2022; Vafa et al., 2024; Spada et al., 2024). In this study, it was used to explore the potential of different novel and commercially available seaweed-based bio-effectors. As plant bio-effectors, they were evaluated for their potential mineral use efficiency, growth-promoting effect and drought resilience in combination with adapted plant nutrition strategies. It was also important to evaluate bio-effectors' contribution to helping the EU farm-to-fork strategy attain its goal of reducing pesticides and mineral fertilisers.

A first filter paper screening tests were first conducted to identify promising bio-effector products and their effective concentrations to improve the germination, root and shoot growth of wheat seedlings. Following this preliminary evaluation, a pot experiment to further test the effectiveness of most promising product concentrations was conducted in a greenhouse at University of Hohenheim. The trial was set using a complete randomized design with bio-effectors as treatment factor comprising eight levels in five replicates: Control, concentration? SPAP4, 0.5 ml. l⁻¹ SPB24P5, 5 ml.l⁻¹ SPB24P5, concentration? SPA+BEM, concentration? SWP4, concentration AMLLK and concentration Ukelp. they are the seaweed extracts. The spring durum wheat variety Beladur was provided by the Wheat Breeding and Research Department, University of Hohenheim.

The study was conducted in collaboration with a non-profit organization, Open Climate Solutions, Switzerland (<https://www.openclimatesolutions.earth/>), which provided the innovative seaweed extracts to be tested in the experiment. This organisation aims to promote sustainable agriculture in Europe and worldwide. In addition to the pot experiment, a large scale on-farm field trial was performed at Gut Wilhelmsthal (<https://gut-wilhelmsthal.de/>), a farm located in the east of Saxony-Anhalt.

1.2. Problem statement

Nowadays, agricultural sectors must simultaneously boost output to feed the increasing world population, improve resource use efficiency and decrease the negative impacts on ecosystems and public health (Rohr et al., 2019). Over the past three decades, some technological innovations have been made to enhance agricultural production systems & resilience by using improved varieties and conventional agrochemicals such as fertilizers and pesticides (Pretty et al., 2014). Thereby, modern agriculture heavily depends on fertilizers and pesticides to maximize crop yields and protect against pests and diseases (Parry & Hawkesford, 2010). The use of bio-effectors, derived from plant growth-promoting microorganisms or natural bioactive compounds, along with other green biotechnological strategies, is considered a promising approach to address the growing challenges of crop production linked to climate change, resource scarcity, and environmental sustainability. These strategies have shown their potential effectiveness, especially in controlled experiments that also shed light on the underlying biological mechanisms. However, achieving consistent results during field applications remains a significant hurdle, highlighting the gap between laboratory research and practical implementation (Colla and Rouphael, 2015; Neumann et al., 2024).

A significant research gap remains in advancing the use of bio-effectors for widespread use by farmers. For effective adoption, essential information must be established, including details on suitable regions, crop types, optimal seasons, and specific organisms along with their appropriate dosages, product formulations and application strategies. Once these factors are determined, this knowledge can be disseminated effectively to the agricultural community, allowing farmers to implement these techniques confidently and accurately (Leaky, 2017). The more comprehensive the information researchers generate, the faster farmers can integrate these practices into their operations, ultimately enhancing agricultural productivity and sustainability.

Statista reported that 195.35 million metric tons of nitrogen fertilisers were used in 2021 from 46.31 million in 1965 for the sake of food security (Statista, 2024). However, inadequate application methods and excessive input of nitrogen fertilizers not only result in low use efficiencies, where only 50 % or even lower of the applied fertilizer N is used effectively, whereas the rest is lost to the environment, causing serious problems, such as eutrophication of water, emissions of greenhouse gases and soil acidification (Ruser et al. 2006; Ye et al. 2022; Govindasamy et al. 2023). Inadequate use of nitrogen fertilizers is an alarming threat to soil health and biodiversity (Kumal et al., 2023, Billen et al., 2013; Tripathi et al., 2020).

1.3. Research Objectives

The aims of this study are described in one aspect:

- Investigate the efficacy of innovative seaweed-based bio-Effectors: Examine the effects, optimal application rates, and adaptability of novel seaweed extracts (SPAP4, SPB24P5, SPA+BEM, SWP4) and commercial products (Ukelp, AMLLK) on wheat, focusing on improving mineral use efficiency and drought stress resilience. That potential will determine if bio effectors can support the EU Farm-to-Fork Strategy 's goal to reduce fertilizer and pesticide use by 50% and 20%, respectively, by 2030.

2. Literature review

2.1 Effect on nutrient use efficiency, plant growth and yield

Agricultural history indicates that the domestication and use of seaweed extracts began in the early 1950s, primarily for enhancing soil properties and increasing crop yields (Calvo et al., 2014; Craigie, 2010). Since then, their application has persisted for decades, with ongoing discoveries highlighting their ability to promote soil health and improve resilience to abiotic stress (Khan et al., 2009; Craigie, 2010).

Seaweed extracts are rich in metabolites and trace elements. (Stirk and van Staden 1996; Sivasankari et al., 2006; Khan et al., 2009; Franzoni et al., 2022; Panda et al., 2012). The impact of seaweed-based bio-effectors on soil and plants was described and attributed to many different properties. Their growth-promoting effect and yield improvement has been attributed to the presence of growth-promoting hormones which are cytokinin, indole acetic acids and gibberellins. These hormones are responsible for cell division, cell elongation and differentiation. Root development caused by bio stimulants helps the plant to absorb water and nutrients (Kapoor et al., 2021). Although the phytohormone levels in seaweed extracts may be insufficient to directly induce significant effects at recommended application rates, their active components, such as polysaccharides, proteins, polyunsaturated fatty acids, pigments, and polyphenols, can modulate innate plant pathways to stimulate the biosynthesis of these hormones (Khan et al., 2009; Chojnacka et al., 2012; Wally et al., 2013). Different modes of actions hence may promote nutrient use efficiency. Growth-promoting hormones like auxins induce the development and elongation of roots. While abscisic acid and ethylene improve resilience to abiotic stresses such as drought, temperature fluctuations, and humidity extremes. Cytokinin contributes by promoting cell division and multiplication, further supporting plant growth and development (Khan et al., 2009). Furthermore, seaweed extracts have been reported to be rich in microorganisms which are responsible for symbiotic relationships with plant roots to supplement nitrogen through nitrogen fixation (Craigie, 2011; Franzoni et al., 2022; Rouphael et al., 2017; Mughunth et al., 2024; Deolu-Ajayi et al., 2022; Hurtado et al., 2020). The seaweed extracts are found to have metabolites which bind to essential minerals for easy absorption and water availability by the plants (EL Boukhari et al., 2020). The seaweed extracts have soluble alginates, protein hydrolysates, flavonoids and other chelating agents like humic acids which complex themselves with minerals and make them available, soil aeration and water holding capacity (khan et al., 2009; Colla et al., 2017; Craigie, 2011).

EL Boukhari et al. (2020) found that extract from the seaweeds *Ascophyllum nodosum*, *Ecklonia maxima* and *Kappaphycus alvarezii* have individually enhanced the increase of N, P, K, Mg, Mn, Ca, Cu, S, Zn and B in different parts of various types of plants. *A. nodosum* has performed well to enhance the absorption of Mn, Cu, Mg S and N in oil seed rape and tomato; K in wheat (Billiard et al., 2014; Stamatiadis et al.; 2015). *Ecclonia maxima* proved to promote the absorption of P and K in the leaf of oil seed and Ca of tomato fruits (Di Stasio et al., 2017; Colla et al., 2017). *K. alvarezii* also enhanced N, P, K and S uptake in grain and straw of wheat and maize (Basavaraja et al., 2018; Rathore et al., 2009).

Franzoni et al. (2022) collected evidence of growth-promoting effects of different seaweed extracts. Application of Kelpak, a commercial seaweed extract rich in *E. maxima* on willow plants resulted in stem elongation and an increase in biomass (Digruher et al.,2018). *A. nodosum* has shown a significant effect on spayed olive trees together with boron supplements. They yielded big olives and good-quality olive oil (Chouliars et al., 2009; Craigie, 2011; Nanda et al.,2022). The vegetative growth and yield parameters were more enhanced in the field of mustard plants that received *E. maxima* treatment (Di Stasio et al.,2018).

2.2 Resilience of abiotic stress

Craigie (201) stated how the plants treated with bio stimulants were stimulated to produce secondary metabolites like polysaccharides, flavonoids and phenolic compounds which help the plant in defense mechanisms against stresses. Their antioxidative character is the key aspect for plants to survive abiotic and biotic stress conditions (Rioux et al., 2007; Bakhoun et al., 2023). The compounds that accumulate during drought stress are called osmolytes and include proline, sorbitol, and glycine betaine. They can protect to plants from stress by adding to osmotic regulation of the stomata cells, protect enzymes, act as antioxidants and scavenge free radicals. This prevents the plant from losing water (Sharma et al., 2019). Charophytes, a group of freshwater green algae, have been found to thrive for different environmental conditions and to be effective in helping plants to withstand harsh conditions. (Zhang et al., 2017, Gitau et al., 2022)

Seaweed-based bio-effectors also have been identified to enhance plant resilience against biotic stress. For instance, the decline in various pathogenic fungal communities and an increase of beneficial microorganisms were observed after seaweed extract application on the soil under tomato plantation (Wang et al., 2018). A foliar spray of seaweed extracts improved the cultivation of baby lettuce in low-nitrogen environments; without a surprise, the Nitrogen content in the leaves increased despite the nitrogen-poor cultivation media (Di Mola *et al.*, 2019b).

Ascophyllum nodosum extract (ANE) is one of the most prominent seaweeds used for the production of extracts for application on crop plants. It is mostly known for its antioxidant character which enhances plants' resilience against biotic stress (Vafa et al., 2024). ANE also came out as an outstanding seaweed extract for leafy vegetables when it prevented the destruction of chloroplast and increased the green colour of broccoli and spinach (Khan et al., 2009; Lora-luz et al., 2014; Franzoni et al., 2022).

3. Materials and Methods

3.1 Preliminary screening test to identify effective product concentrations

One month before the bio-effector application in the pot experiment (1st April 2024), a screening test experiment was set according to Table 5.

For spray application of the algae extracts those concentrations were selected that had shown most promising stimulatory effects on root and shoot elongation of germinating durum (var. Beladur) seeds in a preceding germination test. For this purpose, different concentrations of the algae extracts (0.0 %, 0.5 %, 2.5%, and 5% v/v) were prepared by diluting the extract with deionized water. Then filter paper sheets (Machery-Nagel, Type 710, format 58 cm x 58 cm) were soaked thoroughly with each 50 ml the prepared algae extract dilutions. Ten durum wheat seeds were placed on each filter paper. Folding filter paper sheets into bags (ca. 18.0 x 19.3 cm) and putting them into plastic bags helped to maintain moisture, ensure seed organization, and provide aeration for optimal seedling growth (Figure 1).

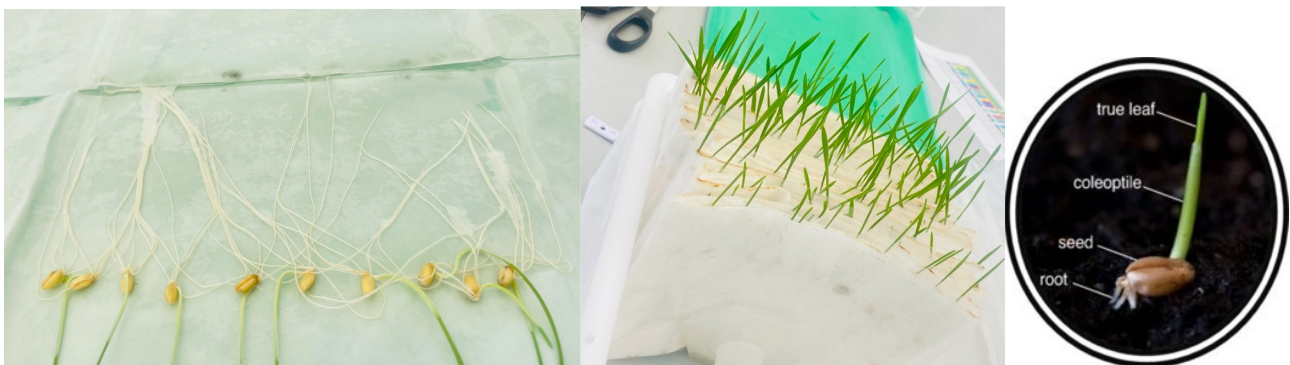
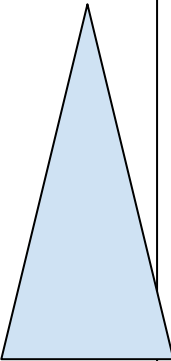


Figure 1: Screening test pictures in filter paper

After incubation for 15 days at room temperature for germination (25°C) the germination rate, root length, the number of root tips per plant, and the length of the coleoptile respectively first leaf was recorded. Results were analyzed by statistical comparison of the growth variables across the different concentrations to assess the phytohormonal effects, such as stimulation or inhibition of

germination and root or shoot growth. This method helped to determine the bioactivity of the algae extracts on seed germination and early seedling growth.

Table 1: The preparation of application solution for a screening test

Treatment	Treatment (concentration)	Preparation of application solution	
Ctrl	only deionized water	only deionized water	
Bio-effector	0.5 ml/liter	take 20 ml (or g) from 2.5SPAP4 solution and add another 80 ml (or g) of water	
Bio-effector	2.5 ml/litre	(take 100 ml (or g) from 5.0SPAP4 solution and add another 100 ml (or g) of water	
Bio-effector	5.0 ml/litre	1.0 ml + 199 ml water	

3.2 Experimental Design

The pot experiment was set-up in a one-factorial complete randomised design in the greenhouse at University of Hohenheim with the spring durum variety Beladur. Test factor were the different levels of bio-effector treatments. Table 2 provides an overview on the different algae products provided by the Swiss non-profit organization Open Climate Solutions including two commercial seaweed extracts that were used as reference products. Regarding the types of algae, the extraction products could be divided into two categories: freshwater and seaweed algae. They were also produced differently; some were processed through fermentation others through wet milling. In the pot experiment 8 different treatment variants, which comprised the untreated Control, three freshwater seaweed extracts and two non-fresh (saline) water seaweed extracts. One of the freshwater seaweed extracts (SPB24P4) was tested in two different concentrations (0.5 ml l⁻¹ and 5 ml l⁻¹). The whole number of experimental units was 40 comprising eight levels of bio-effector treatments in 5 replicates (Figure 2). Each experimental unit occupied one Mitscherlich pot and at least 22 seeds were sown on 26th February 2024.

SPA+BEM	AMLLK	5SPB24P5	Ukelp	0.5SPB24P5	SPAP4	Ctrl	5SPB24P5	AMLLK	SPAP4	SWP4	
0.5SPB24P5	AMLLK	SPAP4	Ukelp	SWP4	SPA+BEM	Ukelp	SWP4	SWP4	SPA+BEM	0.5SPB24P5	SPAP4
Ukelp	Ctrl	Ctrl	SPA+BEM	Ctrl	5SPB24P5	Ctrl	5SPB24P5	0.5SPB24P5	5SPB24P5	AMLLK	

Figure 2: Randomization of treatments

Table 2: The details of Bio-effectors used in the trial

Bio effector	Species	Habitat	Production	Current Status
AMLLK, Atlantik Mariculture Limited, Ardtoe Marine Laboratory, Ardtoe, Acharacle, UK	Seaweed algae (<i>S. latissima</i>)	Non-freshwater seaweed	fermentation	Commercial product
SWP4, Open Climate Solutions			Wet milling	innovative product
SPA+BEM, Open Climate Solutions	Freshwater algae (Charophyta)	Freshwater algae	Fermentation	innovative product
SPB24P4, Open Climate Solutions			Wet milling	innovative product
SPAP4, Open Climate Solutions			Wet milling	innovative product
Ukelp, Abundigro, BK Zeewolde, Netherlands; Liquid KelpTM,	<i>E. maxima</i>	Non-freshwater	Pure cold pressed	Commercial product

3.3 Soil substrate and basic fertilization

3.3.1 Soil substrate

The soil originated from leveling work on topsoil of grassland at the campus of University of Hohenheim near the greenhouse (48°42'46.0"N 9°12'29.0"E) in 2019. The soil type, as commonly found on the Filder plain, is a Parabraunerde, which has high storage capacity for plant-available water and is rich in Ca, Mg, and K (Lehrgarten Agrarwissenschaften, 2023). Table 3 presents physical soil properties of a soil profile typical of Luvisol on the Filder plain and results of soil pH and mineral nutrient extractions analyses performed with the sieved soil charge used for the pot experiment.

Table 3: Chemical and physical characteristics of the topsoil from a Luvisol used for the pot experiment

Soil features	Values/information
Parent rock	loes
Horizon	humic topsoil layer under grassland
soil pH (CaCl ₂)	7.4
Extractable P, K, Mg	P (CAL): 36 mg kg ⁻¹ soil dry matter corresponds to content class C (adequate) K (CAL): 96 mg kg ⁻¹ soil dry matter corresponds to content class C (adequate) Mg (CaCl ₂): 136 mg kg ⁻¹ soil dry matter corresponds to content class C (adequate, K:Mg ratio should be approximately 3:1 for optimal acquisition by the plant, therefore no Mg fertilizer was applied)
Texture	silty clay loam
Maximum water holding capacity (WHC max)	33,2 g water 100 g ⁻¹ soil dry matter

3.3.2 Basic fertilisation

Before mixing with the quartz and filling into the pots, the sieved (5 mm mesh) soil was fertilized by spray application of mineral nutrient solutions that were evenly admixed. The amounts of mineral nutrient solutions to be added to the soil, were calculated based on the soil test results. Ammonium phosphate and calcium nitrate were chosen as fertilizer forms for the nutrient nitrogen. This aims to create a balanced ratio between ammonium and nitrate, as excessively high NH_4^+ concentrations can be toxic to plants (Britto & Kronzucker, 2002), whereas adequate supply can promote root colonization by beneficial microorganisms (Mang et al., 2023) and the acquisition of sparingly available mineral nutrients (Kostic Kravljanac et al. 2024). Nitrate not only acts as a nutrient, but also plays a key role as a signaling molecule, influencing various aspects of plant growth and development (Jia et al. 2023).

The following fertilizer forms were used: Ammonium dihydrogen phosphate $\text{NH}_4(\text{H}_2\text{PO}_4)$ from Merck, calcium nitrate tetrahydrate $\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$ from ChemSolut, and potassium sulfate K_2SO_4 . No addition of mineral magnesium fertilization was required, as 136 mg of Mg kg⁻¹ soil dry matter CaCl_2 -extractable magnesium was already present in the soil (Table 3). The application rates for N, P and K per kg soil dry matter are shown in Table 3.

Table 3: Application rates of N, P and K fertilizers

target values	[mg kg ⁻¹ Soil DM]
N	60
P	80
K	150

3.3.3 Preparation of Nutrient Solutions

The first step was calculating the required amount of nutrient solution to meet the target values. For this purpose, separate preparations were made for each fertilizer.

Potassium Sulfate Solution: A 0.5 molar K₂SO₄ solution was prepared and applied to the soil via spraying. The application provided 150 mg of potassium per kilogram of dry soil mass.

Ammonium Phosphate and Calcium Nitrate Solutions: For fertilization with nitrogen (N) and phosphorus (P), solutions of calcium nitrate and ammonium phosphate were prepared to deliver 10 mg of N and 10 mg of P per gram of solution, respectively. When using ammonium phosphate, the P target value of 80 mg kg⁻¹ dry mass (DM) was first satisfied, and the amount of nitrogen supplied as ammonium was calculated. For determining the required amount of calcium nitrate, the ammonium nitrogen contribution was subtracted from the target nitrogen value.

The required amount of ammonium phosphate per gram of solution was 37.14 mg. This amount contained 4.52 mg of nitrogen. To meet the phosphorus target of 80 mg kg⁻¹ DM, 36.18 mg of nitrogen was provided by ammonium phosphate. For nitrogen fertilization with calcium nitrate, the nitrogen target for low-N (60 mg/kg DM) was reduced by this 36.18 mg to 23.82 mg/kg DM.

Application of fertilizer solutions to the sieved soil: In the laboratory, the nutrient salts were weighed, dissolved in water by shaking, and transferred to spray bottles. The solutions were then sprayed in layers onto the soil in trays. To ensure uniform nutrient distribution, the soil was sieved to break up clumps and then mixed. The fully prepared soil was covered to prevent drying until sowing. Soil for five pots was set aside for soil treatment with BF Soil (see Treatment section). Before sowing, a layer of fleece and a metal disk were placed in the pots to prevent roots from growing into the lower part of the pot and to allow excess water to drain.

After applying the fertilizers dissolved in water, the water content increased from 11.2% to 15%. This value was also determined by drying at 105°C until constant weight.

Soil texture significantly influences water and nutrient retention capacity and is therefore relevant for conducting and evaluating experiments. The texture of the soil used for the pot experiment was

determined using the “finger test” (LTZ Augustenberg Deller & Schmid, 2024). By shaping and rubbing the soil between the thumb and forefinger, its graininess, silkiness, cohesiveness, and stickiness were assessed. From these characteristics, it was concluded that the original soil has a high clay and silt content, which, although capable of storing significant amounts of water, is also prone to crusting and compaction, potentially hindering air exchange, water infiltration and root growth later. To address this, quartz sand with grain sizes of 0.6–1.2 mm (Type 7) was mixed with the soil ratio of 70 % soil and 30 % sand on dry matter basis. For a total weight of 5 kg per pot, this corresponds to 3.5 kg of soil dry matter and 1.5 kg of sand per pot. Taking into account a soil moisture content of 15% after admixture of fertilizer solutions, each pot was filled with 4.025 kg of fresh soil and 1.5 kg of sand, which means in total 5.525 kg of freshly fertilized soil substrate. To prevent the formation of gaps, the soil substrate was gently compacted after being filled into the pots. The soil containers self-draining pots called Mitscherlich pots were used.

Sowing of durum wheat seeds: Based on the surface area of the Mitscherlich pots (314.14 cm²) and a typical field sowing density of 400 viable seeds/m², a sowing rate of 12.58 seeds per pot was calculated. However, for a pot experiment, higher sowing densities are possible due to the expected lower tillering. Therefore, we decided on a sowing rate of 22 seeds per pot. Sowing took place on 26. February 2024. In slightly moistened and compacted substrate, 22 holes, each 2.5 cm deep, were made at uniform intervals using a rod. One seed was placed in each hole, covered with substrate, and gently pressed down.

The surface was then covered with a 2.5 cm layer of washed and redried quartz sand to prevent soil crusting and erosion. For initial watering, 300 ml of water was applied twice, achieving a water content of 23.2% (see next chapter).



Figure 3: Sowing of durum wheat seeds in Mitscherlich pots

3.3.4 Soil moisture and watering

The watering of the plants followed the calculated maximum and minimum moisture content of each pot. The pot was kept moist above the minimum moisture content and below the maximum moisture content. The drought phase period was induced to enhance the expression of beneficial bio-effector traits in stress mitigation. During drought, they experienced the moisture content below the minimum threshold of the preceding well-watered phase for 23 days from 7th May to 1st June 2024. The total water consumed by each pot from sowing of the wheat seeds until experimental harvest of the plants was recorded and included the moisture added by spray application of the bio-effector products.

3.3.5 Application of the algae extracts

For the pot experiment, the seaweed extract solution was prepared based on the screening tests for each treatment. The concentration that showed a significant difference compared to the control (0%, deionized water only) and performed well across two of the tested parameters was selected. For bio-effectors with no significant differences, the concentration with the highest mean value across two parameters was chosen, as presented in Table 4.

Table 4 : Selection of the concentration from screening test results (Conc. with the Highest Mean)

Bio-effector	Coleoptile length (ml.l ⁻¹)	Root length (ml.l ⁻¹)	Root tip number (ml.l ⁻¹)	Selected Concentration (ml.l ⁻¹)	Exception
Control	De-ionised water	De-ionised water	De-ionised water	De-ionised water	
SPAP4	0.5	0.5	0.5	0.5	
SPB24P5	0.5	0.5	0.5	0.5 & 2.5 (exception)	Both are meant to be evaluated
SPA+BEM	5.0	0.5	5.0	5.0	
SWP4	0.5	5.0	2.5	5.0	recommended
AMLLK	5.0	5.0	2.5	5.0	Recommended by Jensen & Jorgensen, (2022)
Ukelp	5.0	0.5	2.5	5.0	manufacturer recommend

The foliar application of seaweed extracts was done twice. The first spray was on 22nd April 2024 in normal watering time. The second spray was done on 21st May 2024 at the start after the induction of the drought phase. The solutions were prepared and selected contractions were respected.



Figure 4: Pots of the same treatment variant isolated for foliar spraying

Growth parameters: During the pot experiment, the growth parameters were recorded. Plant height was measured from the ground by using meter labelled stick and the chlorophyll content was measured using a SPAD meter. During this growing period, the water consumption data were recorded on every watering day. The moisture was supposed to be kept between 17%- 23% moisture content of the pot's media.

Harvesting data: during harvesting, the flesh and dry weight of above and belowground biomass were recorded using an electronic balance. Green leaves, necrotic parts, stem and head of wheat were weighed apart.

Nutrient analysis: After drying the biomass, the milling was done, and parts were milled together. heads and healthy leaves were milled together. Then roots, necrotic leaves and stems were also milled together. the two types of samples were taken at Hohenheim laboratory for Nutrient analysis. The microwave extraction was done, and the samples were ready for further analysis. Phosphorus was analysed using a photometer, magnesium and Zn were tested by Atomic Absorption Spectroscopy (AAS) and Calcium and Potassium were tested using Flame Atomic Absorption Spectroscopy. The procedure was done following the VDLUFA protocol.

Large scale field experiment with winter wheat

The parallel field experiment with winter wheat sown was set as a randomised block design at Wilhelmsthal farm, Saxony-Anhalt.

The farm trial tested 2 commercial products TopHealthPlants (The Seaweed Company, CK Schiedam, Netherlands) and Ukelp (Abundigro, BK Zeewolde, Netherlands) that was also tested in the pot experiment. The field trial was conducted on 26.44 hectares of arable land, with 8.36 hectares allocated to the control variant, 9.37 hectares to the Ukelp variant, and 8.71 hectares to the TopHealth Plants variant. The trial was designed in a block layout. Each variant had four repetitions, which are marked as "Trial Site (Ts)" on the field plan (Figure 5). Each of the four blocks contained one repetition of each variant. Winter var. Emerick, KWS wheat was sown on 22. September 2023, and sprayed with the manufacturer-recommended amounts of seaweed extracts on 1. March 2024 (during the tillering phase, 20 weeks after sowing), and again in May. The application rates were as follows: 3 L Ukelp with 150 L water per hectare; 5 L TopHealth with 150 L water per hectare. A crop sprayer with a 36 meter boom was used for the treatment (Figure 5), corresponding to the width of one trial site repetition. The site is located on sandy loam soil.

Across the blocks, there could be several environmental effects observed: the soil become sandier toward the east, and the previous crop (PC) changes along the east-west axis. In the west (Block I), the previous crop was winter rapeseed (WR), whereas in the east (Blocks II-IV), it was winter wheat (WW). The vertical red line on the field plan (Figure 5) marks the transition of the previous crop from winter rapeseed to winter wheat. Additionally, the field plan shows a hatched area in the northern part where water accumulation was observed in March 2024. This area was excluded from scoring and sampling because it differed significantly from the rest and would have skewed the results. The adjacent trial sites 1 and 2 together form Block I for Ukelp; however, only TS2 was measured and sampled.

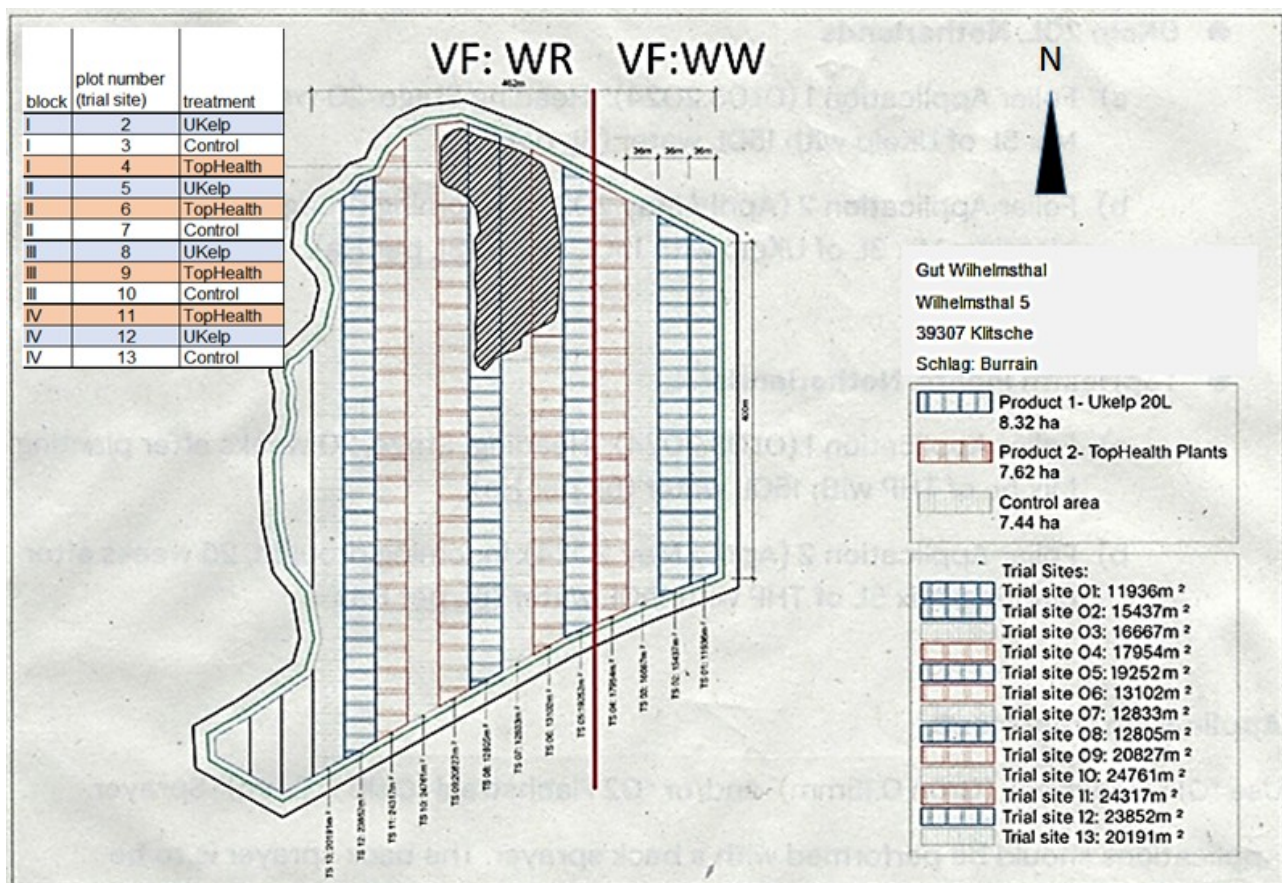


Figure 5: Experimental plan at Wilhelmsthal farm, Saxony Anhalt



Figure 6: Spray application of in the field experiment at Gut Wilhelmsthal, 1. March 2024

Figure 7 illustrates the weather conditions at the next weather station at Genthin near the trial field (11 km away), showing significant rainfall in spring and a drought-prone condition especially in early spring (before mid of May) and summer, apart from a few heavy rain events.

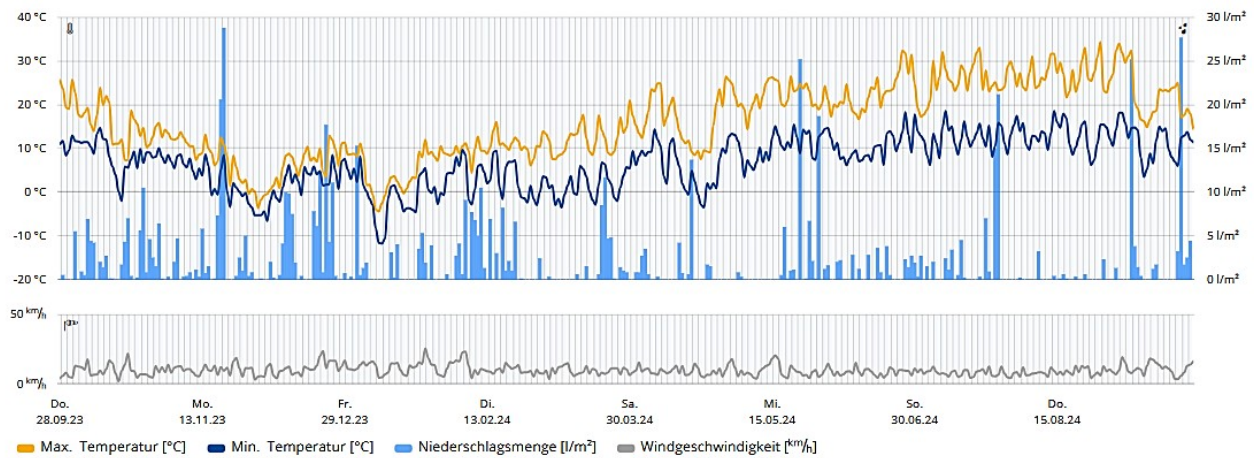


Figure 7: Weather trends during the growth period from October to August. Data from a weather station in Genthin (11 km from Klitsche).

Shortly before the harvest of the field experiment, we visited the farm to discuss the progress of the trial, conduct assessments, and collect plant samples on 12. July 2024 (Figure 8). Within each trial site, plant height was measured at 10 positions (in cm), and the number of ear-bearing stems was counted in eight sections of 1-meter row length. Additionally, five 50 cm row sections per trial site were sampled to determine the aboveground plant biomass and the number of ears per sample. During the harvest in the last week of July, the grain yield of each trial site was recorded by the harvest combine with GPS tracking system.



Figure 8: Collection of plant samples in the field experiment with winter wheat at Gut Wilhemsthal on 12. July 2024

3.4 Statistical analysis

Recorded data from screening test, pot experiment and field experiment were analysed using the linear model for a randomized complete factorial design using the Statistical Analysis System software (SAS) version 9,4 and R software. When the analyses of variance (ANOVAs) indicated significant differences, post hoc tests (LSD test, T-test, and Duncan test) or multiple t-tests were carried out to check the significant difference between the treatments (Mudra, 1958). For Wilhelmsthal data analysis, the complete randomized block design was considered to analyze the block effect.

4. Results

4.1 Screening Test

The results of the screening tests for durum wheat seeds treated with innovative and commercial bio-effectors at concentrations of $0.5 \text{ mL} \cdot \text{L}^{-1}$, $2.5 \text{ mL} \cdot \text{L}^{-1}$, and $5.0 \text{ mL} \cdot \text{L}^{-1}$ are shown in Figure 1. The parameters recorded included coleoptile length, root length, and the number of root tips per plant. Some bio-effectors illustrated a significant growth-promoting effect for coleoptile length compared to the control. The treatment SPA+BEM at $5.0 \text{ mL} \cdot \text{L}^{-1}$ resulted in a coleoptile length of 9.24 cm, the highest observed in this category. SWP4 had effects at $0.5 \text{ mL} \cdot \text{L}^{-1}$ and $5.0 \text{ mL} \cdot \text{L}^{-1}$, producing coleoptile lengths of 8.38 cm and 8.12 cm, respectively, with the lower concentration yielding better results. SPAP4 at $0.5 \text{ mL} \cdot \text{L}^{-1}$ recorded the highest coleoptile length at 7.24 cm. In contrast, Ukelp and AMLLK significantly improved coleoptile length at $5.0 \text{ mL} \cdot \text{L}^{-1}$ concentration, reaching 8.12 cm and 7.29 cm respectively. However, SPB24P5 did not show significant differences from the control at any concentration tested. For root length, no bio-effector treatments caused significant differences compared to the control, regardless of the concentration applied. Similarly, the number of root tips per plant showed no significant variations across treatments or concentrations. In conclusion, the tested bio-effectors exhibited notable effects on coleoptile length but did not significantly influence root length or the number of root tips under the experimental conditions.

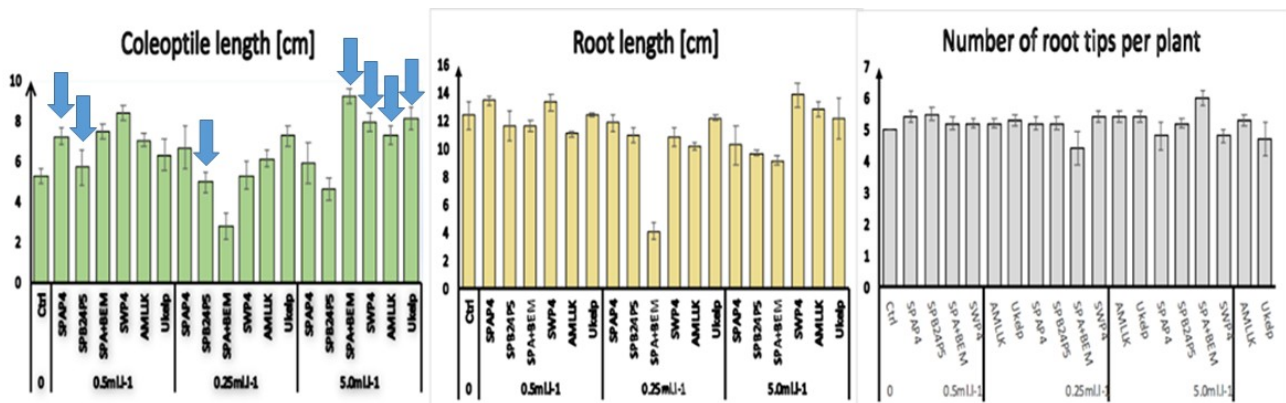


Figure 9: Screening test results. The blue arrows indicate which concentrations were collected for the application in the pot experiment.

4.2 Pot Experiment

4.2.1 Plant height

During the vegetative growth of durum wheat in the pot experiment, plant height was recorded at 13, 23, and 43 days after the first application of bio-effectors. Statistical analysis revealed no significant differences ($P \leq 0.05$) in plant height either among the bio-effector treatments nor between the treatments and the control. Although plant height consistently increased over time, applying bio-effectors did not lead to statistically significant variations. At 13 days after the first application, plant heights ranged narrowly between 42 cm and 44 cm, with all treatments showing comparable growth. By 23 days, heights had increased to between 57 and 59 cm, with similarly small differences among treatments. At 43 days, the heights further increased to a range of 59 cm to 62 cm, still without statistical significance.

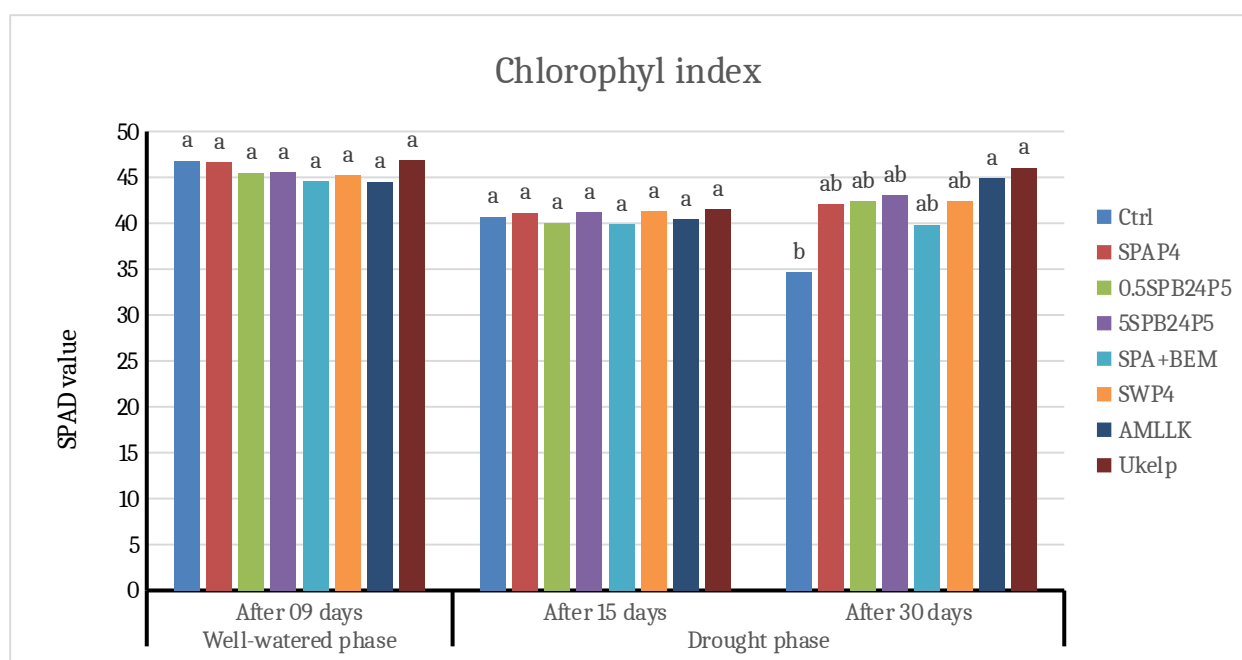


Figure 10: Plant height records at 3 different days after 1st application of bio-effectors.

4.2.2 Water consumption

The experimental design involved maintaining indoor potted plants under controlled watering conditions to monitor water consumption and drought response. Soil moisture was kept within set limits, with pot weights recorded before and after watering to measure consumption. Watering continued for 13 days after the first bio-effector application.

Then, a drought phase was introduced to stimulate the expression of beneficial bio-effector traits for improved stress resilience of the plants. This drought phase lasted for 17 days without any watering. Afterward, pots were watered to maximum soil moisture, and the pot weight difference before and after the drought phase was recorded to assess water consumption. During the well-watered phase, no significant differences in water consumption were observed across treatments. However, by the end of the drought phase, clear distinctions emerged.

The control treatment exhibited the highest water consumption, with a recorded value of 924.8 g of irrigated water per pot. This was significantly higher compared to other treatments ($p \leq 0.05$). Among all treatments, SPA+BEM, AMLLK, and Ukelp showed the significantly lowest water consumption at 762.2 g, 761.0 g, and 760.0 g, respectively ($p \leq 0.05$). The remaining treatments showed no significant differences from the highest or lowest consumption groups.

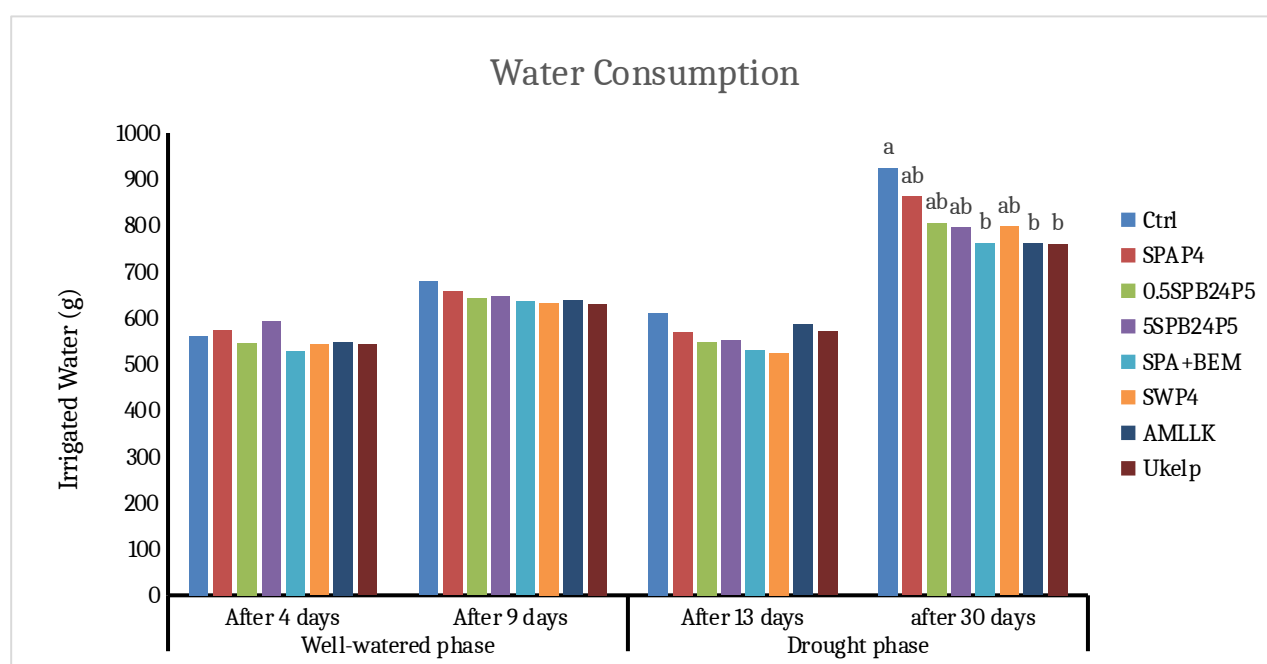


Figure 11: The Water consumption graph at different days after 1st bio-effector application. After 13 days, the drought phase was induced until 30th day. The treatments sharing same letters in superscripts are not significantly different ($p \leq 0.05$).

4.2.3 Chlorophyll index

The study evaluated the effect of bio-effector applications on SPAD values, an indicator of chlorophyll content, under normal and drought conditions. During the normal watering phase, no significant differences in SPAD values were observed across treatments ($P \leq 0.05$) at 9 days after the initial bio-effector application. Similarly, at the beginning of the drought phase, 15 days after the first bio-effector application, SPAD values across treatments exhibited minimal, non-significant variation. However, by the conclusion of the drought phase, clear distinctions in SPAD values emerged among the treatments. The control treatment displayed a marked decrease in SPAD values, reaching the lowest value of 34.65 ($P \leq 0.05$). In contrast, Ukelp and AMLLK treatments maintained significantly higher SPAD values under drought stress, achieving the highest values of 46.05 and 44.92, respectively. Other treatments displayed varied SPAD measurements, but their values were not significantly different from either the highest (Ukelp, AMLLK) or the lowest (control) recorded SPAD values.

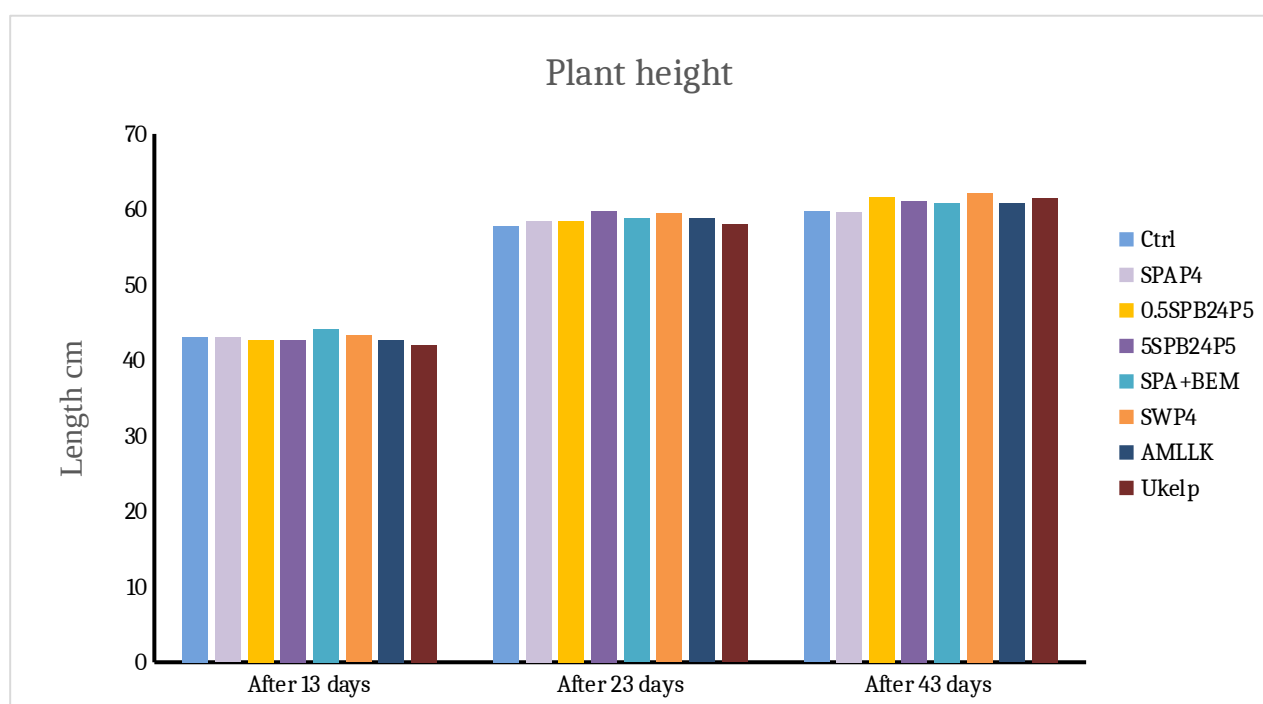


Figure 12: Effect of bio-effectors application on leaf chlorophyll index (SPAD values) of Durum wheat in a pot experiment. X-axis: days after first bio-effector application (9th, 15th and 30th). The treatments with the same letters are not significantly different ($P \leq 0.05$).

4.2.4 Dry matter of roots, shoots and Ears

The dry weights of roots, shoots and ears were analyzed after harvesting to assess the effects of bio-effector treatments on biomass production. The results indicated that pots treated with bio-effectors and the control variant generally produced similar results across all treatment. The root biomass varied across treatments, with the highest mean value observed in response to the treatment with 5ml l⁻¹ SPB24P5 (3.54 g), indicating a stimulatory effect on root growth. The lowest root biomass was recorded for AMLLK (2.21 g), which was lower than the control (3.43 g). Ctrl, SPAP4, and SWP4 showed similar root biomasses with mean values around 3.4 g. 5ml l⁻¹ SPB24P5 recorded the highest mean value (12.79 g), while AMLLK had the lowest (11.64 g). The control group demonstrated intermediate performance (12.38 g). The biomass of ears varied noticeably among treatments. 0.5ml.l⁻¹ SPB24P5 exhibited the highest mean value (2.37 g). In contrast, SPA+BEM had the lowest ear biomass (1.66 g), which was lower than the control (2.06 g).

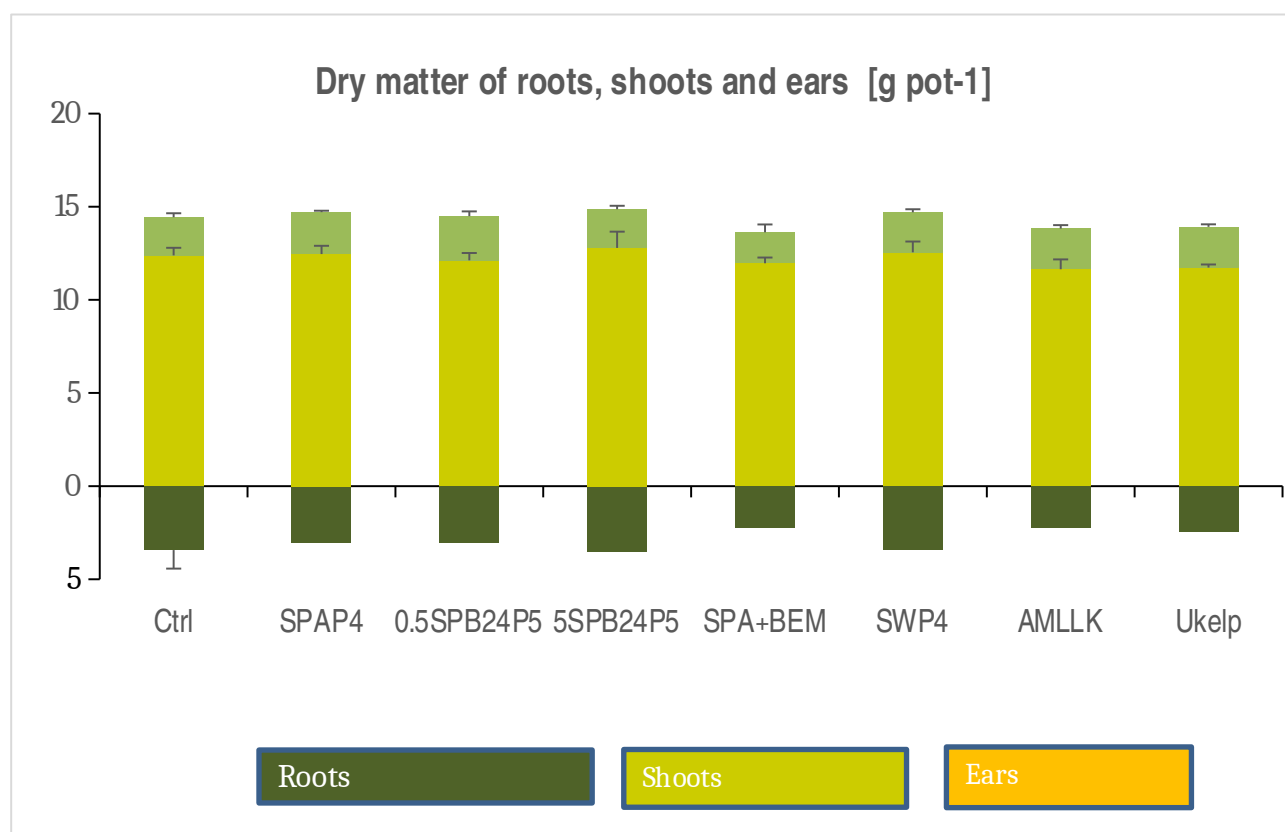


Figure 13: Root, shoot and ear dry biomass at experimental harvest 10 months and 20 days after sowing. The means were not significantly different ($p > 0.05$).

4.2.5 Nutrient analysis

4.2.5.1 Nutrient analysis for whole biomass

The results presented in Figure 6 summarize the effect of bio-effector treatments on the mineral content in whole dry biomass of durum wheat, expressed in mg pot^{-1} . The analysis results scatter the plant tissues that at experimental harvest were still green (Green leaves and ears) from the other (necrotic) plant parts (roots, stem and necrotic leaves). The analysed minerals included potassium (K), phosphorus (P), calcium (Ca), magnesium (Mg), and zinc (Zn). Applying bio-effectors did not lead to any statistically significant differences of the mineral content in either necrotic or green plant parts ($p \leq 0.05$). However, Potassium (K) contents were highest in necrotic part for the control (Ctrl) treatment ($444.92 \text{ mg} \cdot \text{pot}^{-1}$), followed by SWP4 ($428.31 \text{ mg} \cdot \text{pot}^{-1}$). SWP4 exhibited the highest K content in green plant part ($94.02 \text{ mg} \cdot \text{pot}^{-1}$). AMLLK recorded the lowest K content ($324.86 \text{ mg} \cdot \text{pot}^{-1}$) in necrotic plant parts. Calcium (Ca) levels varied among treatments. Necrotic plant parts under $5 \text{ ml} \cdot \text{l}^{-1}$ SPB24P5 treatment had the highest Ca content ($65.71 \text{ mg} \cdot \text{pot}^{-1}$), while the control group recorded the highest Ca content ($63.34 \text{ mg} \cdot \text{pot}^{-1}$) in green plant part. Ukelp displayed the lowest Ca content ($41.51 \text{ mg} \cdot \text{pot}^{-1}$) in necrotic plant parts. For magnesium (Mg), the control group (Ctrl) and SWP4 had the highest contents in necrotic plant parts (21.79 and $21.06 \text{ mg} \cdot \text{pot}^{-1}$, respectively) compared to other treatments. Green plant parts showed the greatest Mg content under the $0.5 \text{ ml} \cdot \text{l}^{-1}$ SPB24P5 treatment ($9.16 \text{ mg} \cdot \text{pot}^{-1}$). Conversely, Ukelp recorded the lowest Mg content in necrotic parts ($17.38 \text{ mg} \cdot \text{pot}^{-1}$). Zinc (Zn) content was relatively consistent across treatments, with $5 \text{ ml} \cdot \text{l}^{-1}$ SPB24P5 exhibiting the highest Zn levels in necrotic part ($0.42 \text{ mg} \cdot \text{pot}^{-1}$). Zn contents in green plant parts were highest in SPAP4 ($0.19 \text{ mg} \cdot \text{pot}^{-1}$). SPA+BEM had the lowest Zn content ($0.39 \text{ mg} \cdot \text{pot}^{-1}$) in necrotic plant parts. Phosphorus (P) content was notably higher in necrotic part under the control (Ctrl) treatment ($44.19 \text{ mg} \cdot \text{pot}^{-1}$) and $5 \text{ ml} \cdot \text{l}^{-1}$ SPB24P5 ($41.73 \text{ mg} \cdot \text{pot}^{-1}$). Green parts displayed the highest P levels under $0.5 \text{ ml} \cdot \text{l}^{-1}$ SPB24P5 ($18.96 \text{ mg} \cdot \text{pot}^{-1}$). AMLLK recorded the lowest P contents ($33.26 \text{ mg} \cdot \text{pot}^{-1}$) in necrotic plant parts.

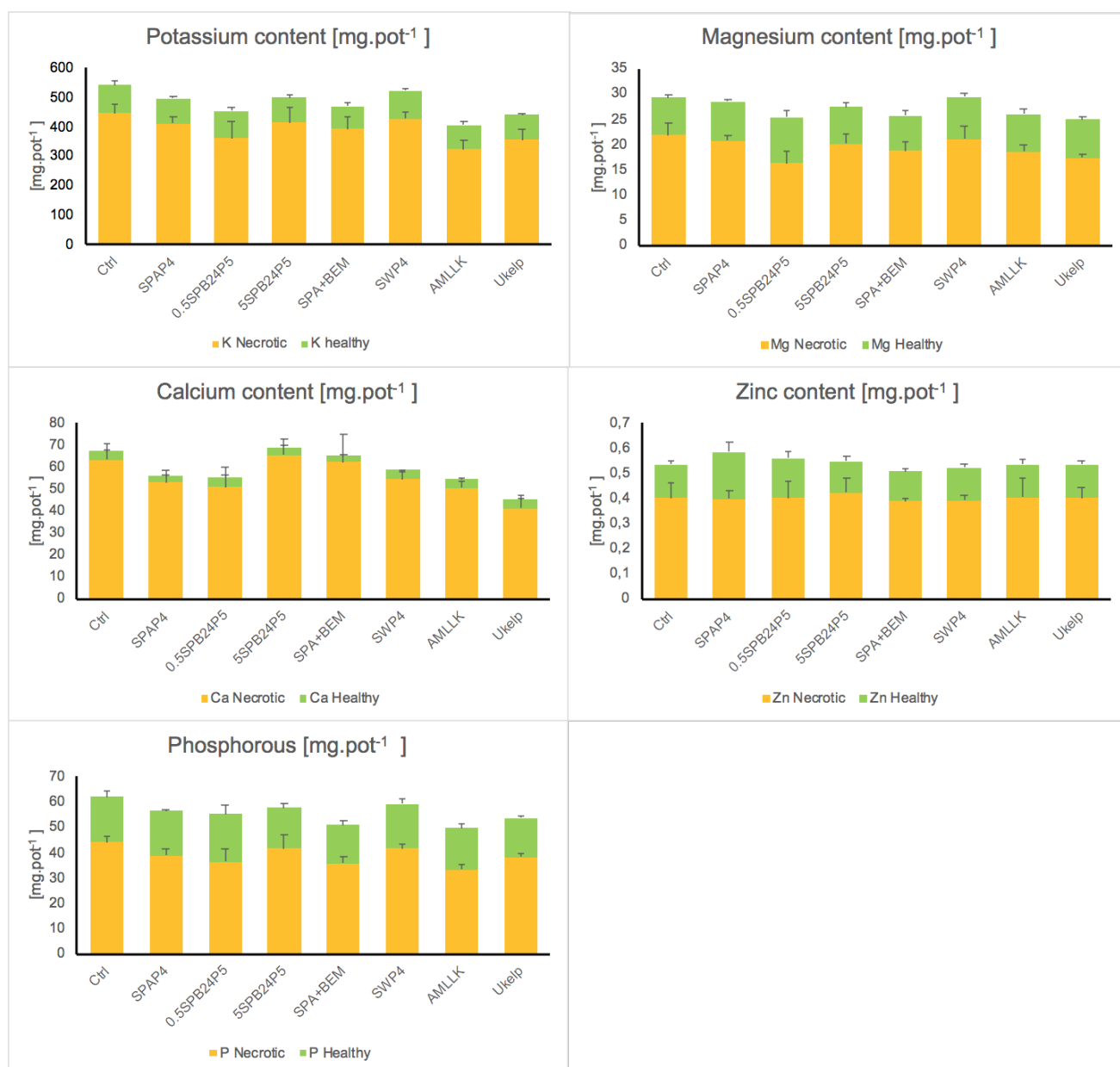


Figure 14: Contents of mineral nutrients in healthy part (green leaves and ears) and necrotic part (roots, stems and necrotic leaves) of durum wheat at harvest of the pot experiment

4.2.5.1 Nutrient Concentrations in green plant parts (healthy leaves and ears)

The results presented in Table 7 summarize the concentrations of mineral nutrients the plant parts of durum wheat that at experimental harvest were still green. The examined minerals included potassium (K), phosphorus (P), calcium (Ca), magnesium (Mg), and zinc (Zn). Statistical analysis indicated a significant difference ($p \leq 0.05$) across treatments for most mineral concentrations, as denoted by mean values not sharing same letters in subscripts. For calcium concentration, the AMLLK treatment showed the highest value ($1.15 \text{ g Ca kg}^{-1}$ dry matter), whereas SPAP4 had the lowest value (0.89 g.kg^{-1}). Magnesium concentration ranged narrowly, from 1.96 g.kg^{-1} (SPAP4) to 2.17 g.kg^{-1} (0.5 ml.l^{-1} SPB24P5), which were significantly different from each other, but not from

the control and other treatments . Zinc concentration also exhibited minimal variation, with SPAP4 recording the highest value (47 mg.kg⁻¹), followed by AMLLK (37 mg.kg⁻¹), while SWP4 had the lowest zinc content (32 mg.kg⁻¹). Potassium concentrations ranged from 21.71 g kg⁻¹ in the SPAP4 treatment to 25.76 g kg⁻¹ in the control. Although numerically higher in the control, this difference was not statistically significant. Phosphorus levels followed a similar trend, with the control showing the highest value (4.89 g.kg⁻¹), while the lowest value (4.28 g.kg⁻¹) was observed in the Ukelp treatment. These variations, however, were not statistically significant.

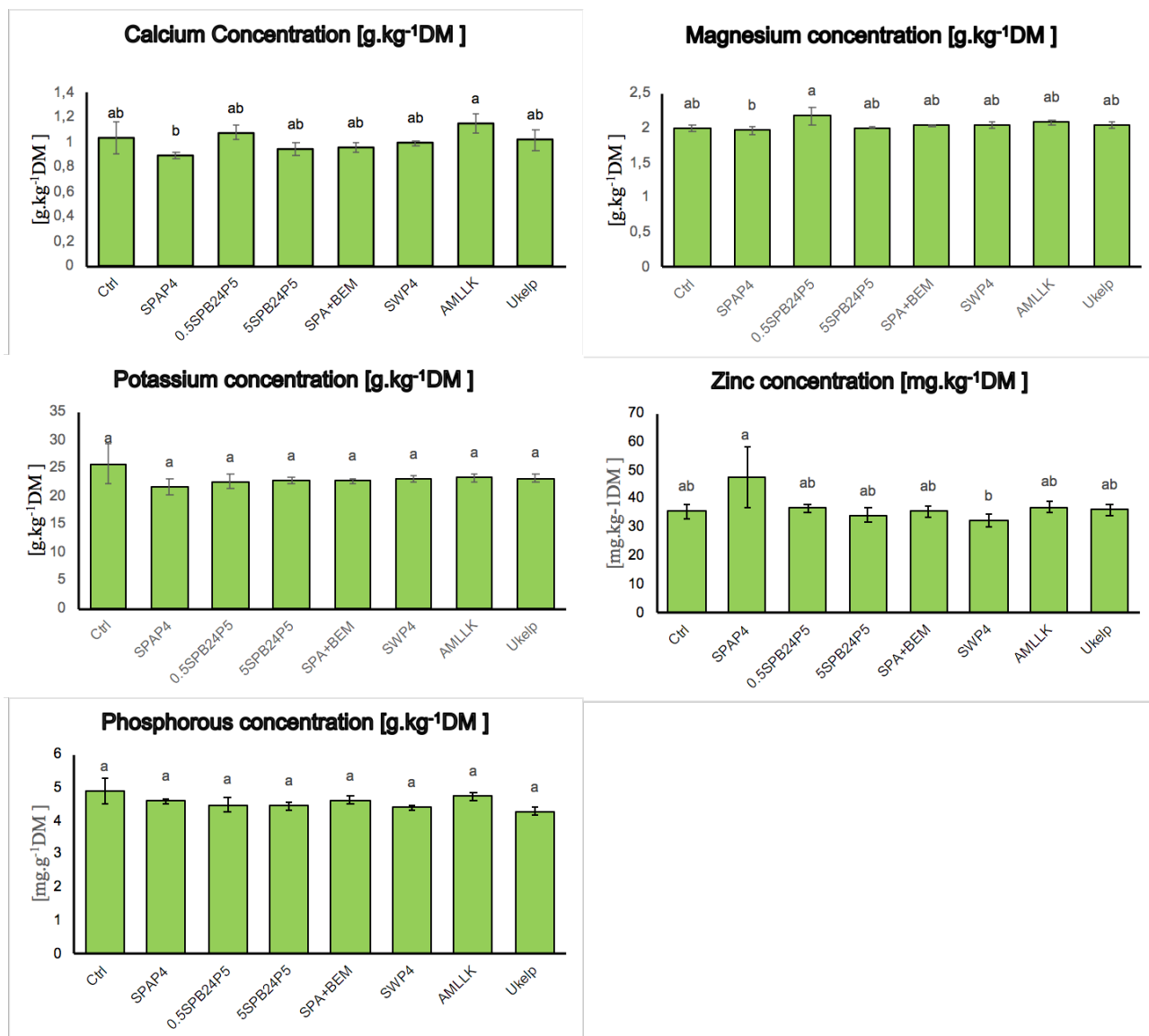


Figure 15: Average mineral concentrations in green plant parts (green leaves and ears) of durum wheat in pot experiment (treatments sharing same letters in subscripts are not significantly different ($p \leq 0.05$)).

4.3 Results of the field experiment at Gut Wilhelmstal

4.3.1 Plant growth

The effect of bio-effectors on the plant height, ear number and dry mass of winter wheat at Gut Wilhelmstal farm showed almost no statistically significant differences between treatments as illustrated in Figure 6. In contrast, the block factors revealed a significant difference ($p \leq 0.05$). Plant height was nearly identical across treatments, with TopHealth, control, and Ukelp achieving heights of 95.0 cm, 94.5 cm, and 93.3 cm, respectively. The ranking of ear number differed slightly, with TopHealth, Ukelp, and control recording 91.7, 88.7, and 86.8 ears per meter row length, respectively. However, total above ground dry mass revealed a significant difference between TopHealth (16.21 t ha⁻¹) and Ukelp (14.25 t ha⁻¹), though neither was significantly different from the control (15.60 t ha⁻¹).

Analysis of the block effect showed that the plant height in blocks IV, III and II had consistent growth without a significant difference, but the plants in block I showed significantly lower plant height. These results were similar in plant height, ear number and dry mass. block IV, III and II had similar plant heights of 97.6, 97.3 and 95.7 cm respectively while block I had the lowest plant height of 86.5 cm. Similarly, the number of ears was affected by the block effect only in block I which had the lowest number of 76.3 ears m⁻² while blocks VI, III and II were not significantly different from each other with 92.2, 96.1 and 91.1 ears m⁻² respectively. Again, Block I in dry mass attended the lowest weight of 14.08 t ha⁻¹ while Block IV, III and II showed no significant difference from each other with 16.1, 16.2 and 15.2 t ha⁻¹ respectively.

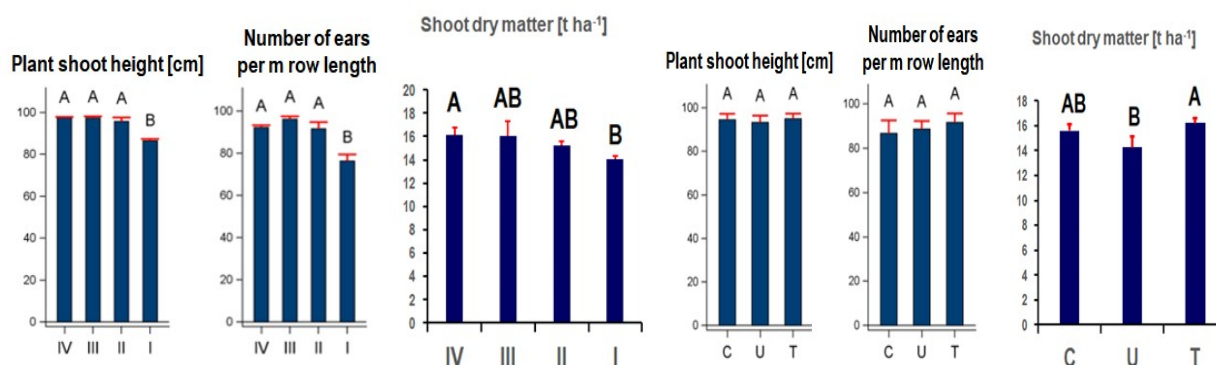


Figure 16: Analysis of plant height, ear number and dry mass for block and treatment factors effects ect (Blocks or treatments sharing same letters in superscripts are not significantly different ($P \leq 0.05$)).

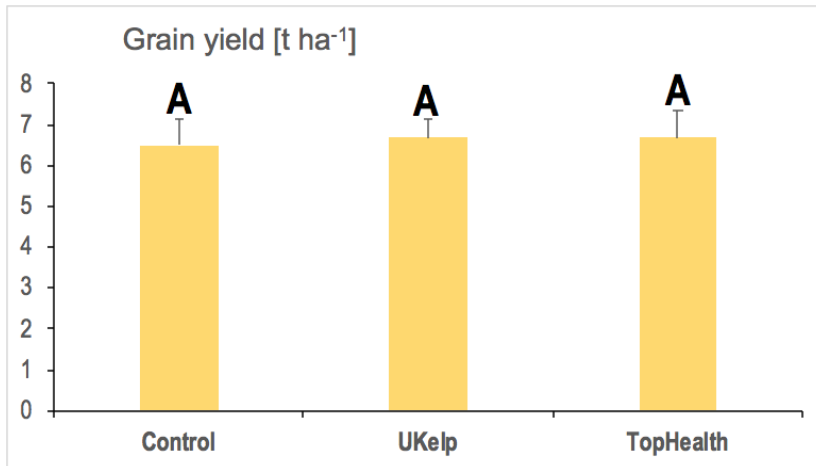


Figure 17: Grain yield of winter wheat at Gut Whilhemstal.

The grain yield results obtained from harvest combine with GPSD tracking revealed that there is no significant difference in grain yield of all treatments. The Ukelp and TopHealth had the same yield of 6,67 t. ha⁻¹ and the control had less grain yield of 6,49 t. ha⁻¹.

Figure 19: illustrates the analysis of the above-ground biomass of winter wheat at Gut Whilhemstal. The results straw and ears biomass of the wheat showed a significant difference between TopHealth and Ukelp. Similar to the grain yield, the control was not significantly different to either TopHealth or Ukelp in straw biomass and ears. TopHealth attended highest 6.61 and 9.71 t. ha⁻¹ of straw and ears dry weight respectively.

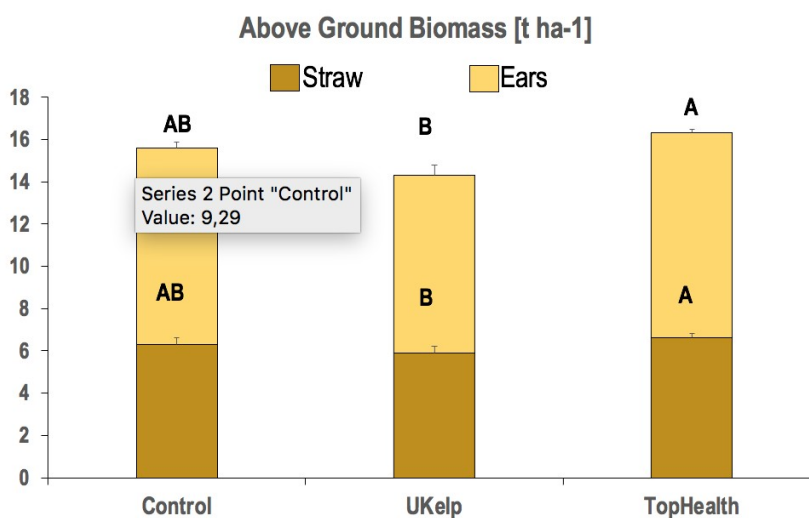


Figure 18: Analysis of above-ground biomass at Gut Wilhelmstal

4.3.2 Field experiment Nutrient test results

The nutrient analysis of wheat grain harvested at Gut Wilhelmstal was done to determine phosphorus and potassium concentrations and removal with grain harvest. The results showed that there was no statistically significant difference between treatments regarding phosphorus (3.3 g P kg⁻¹ dry matter) and potassium (4.5 g K kg⁻¹ dry matter) concentrations in wheat grains as well as regarding phosphorus (19 kg P ha⁻¹) and potassium (26 kg ha⁻¹) removal (concentration in grains * grain yield ha⁻¹) with grain harvest ($p > 0.05$). (if there are only marginal differences outlined explanation of mean values makes no sense. Just mention the overall mean as indicated above)

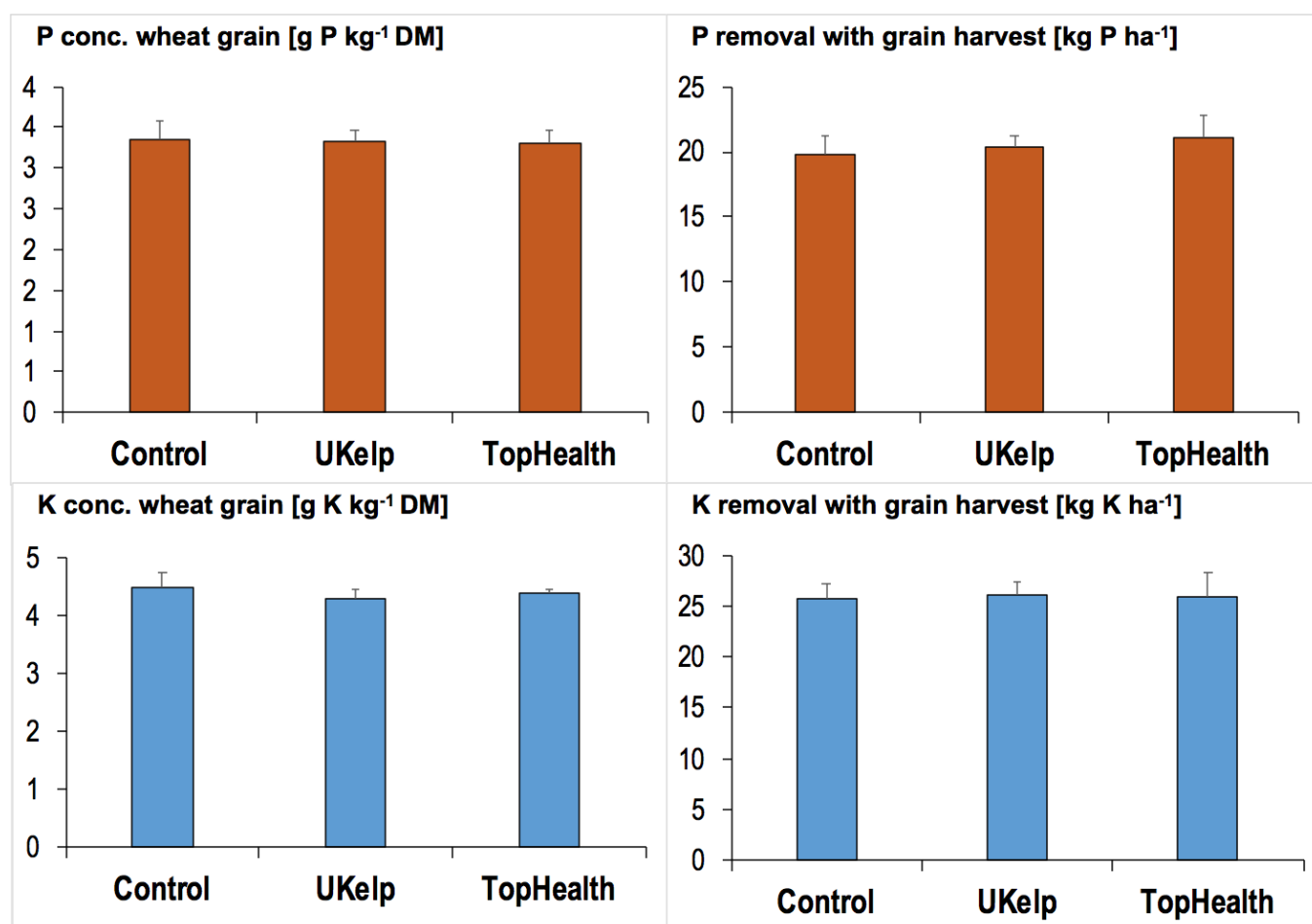


Figure 19: Analysis of grain Nutrient concentration and nutrient removal with winter wheat grain harvest at Gut Wilhelmstal.

5. Discussion

Brown seaweed extracts are one of the recognized bio stimulants for enhancing nutrient use efficiency, plant growth, yield, and resilience to various stress factors (Pacheco et al., 202; Khan et al., 2009; Craigie, 2010). This is largely attributed to the presense of plant growth-promoting phytohormones and secondary metabolites to strengthen plant defense mechanisms (Rioux et al. 2007; Bakhoum et al.,2023). Previous studies suggest that the effectiveness of seaweed extracts is even more significant when applied in low concentrations (Khan et al., 2009; Bhattacharya et al., 2015). Additionally, the response to bio-effectors can be influenced by the genetic makeup of the plant species or interactions with the plant and soil microbiome, further complicating the assessment of their impact (Boutahiri et al., 2024). Our study evaluated both innovative seaweed extracts derived from Swiss seaweed extracts and commercial seaweed products widely available. While commercial products are well-documented, more information on the novel seaweeds extracts is needed. The pot experiments were conducted using fertile garden soil, which may have moderated the effects of the seaweed extracts, as poorer soils often amplify their impact (Li et al 2022). Furthermore, an induced drought phase during the experiment elicited notable responses in terms of decreased water consumption and improved maintenance of leaf chlorophyll contents as assessed by SPAD measurements from the bio-effectors, highlighting their potential effectiveness under moderate stress conditions (De clerqc et al., 2023; Di stasio et al., 2018).

5.1 Plant growth parameters and abiotic stress resilience

Screening test and plant height

A screening test was performed before applying bio-effectors to evaluate their potential for growth promotion. The initial estimation suggested that bio-effectors would exhibit growth-promoting effects, particularly at low concentrations. The results indicated a positive growth-promoting effect, although no statistically significant differences were observed between bio-effector treatments and the control. The concentration effects varied among different bio-effectors, and the results did not confirm that the lowest concentrations were consistently effective. Therefore, concentrations with higher effects at each bio-effector were applied to evaluate their impact on growth parameters, such as coleoptile and root lengths.

It was hypothesized that bio-effector-treated plants would show enhanced root length, coleoptile length, and an increased number of root tips compared to control plants. This expectation was based on evidence suggesting that seaweed extracts contain active ingredients with plant growth-

promoting properties mimicking the effects of phytohormones like auxins responsible for root cell elongation and differentiation (Khan et al., 2009; Craigie, 2010). Additionally, the presence of secondary metabolites in seaweed extracts was anticipated to improve mineral and water absorption, promoting quicker development of coleoptiles and shoots (Rioux et al. 2007; Bakhoun et al., 2023; EL Boukhari et al., 2020). However, despite from promising effects observed in the screening test, the plant height of durum wheat in pot and field trial did not differ significantly among bio-effector treatments, nor between the treated and control groups under normal watering or drought stress conditions. Similarly, it was in agreement with the field experiments at Wilhelmstal. These findings align with Szczepanek et al. (2018), who reported the lack of significant difference in tiller length, number of branches, root weight and grain yield in spring wheat treated Kelpak (*Ecklonia maxima*) seaweed extracts with single application. However, sequential application improved the abovementioned traits. (Muhamad et al., 2013). Shibaeva et al. (2023), have found that the higher concentrations (1200 mg. l⁻¹) of seaweed extract (*Latissima saccharina*) decreased the growth rate and biomass accumulation in wheat plants but a significant difference was detected at lower concentrations (60, 120 and 300 mg. l⁻¹)

Chlorophyll Content

The application of algae extracts notably influenced chlorophyll content, particularly during drought stress. Studies have shown that seaweed extracts can enhance photosynthetic pigments, including chlorophyll content (Spada et al., 2024; Spinelli et al., 2010; Blunden et al., 1997; Fan et al., 2013; Jannin et al., 2013; Mancuso et al., 2006; Sivasankari et al., 2006). In this study, algae extract-treated plants demonstrated increased chlorophyll content during drought stress. Specifically, Ukelp and AMLLK treatments showed maintained SPAD values under drought stress. This indicates that while Ukelp and AMLLK treatments effectively enhanced chlorophyll retention under drought stress, other bio-effectors had intermediate effects, which were not statistically distinct from the extremes. This suggests that Ukelp and AMLLK may delay senescence and protect chloroplasts during drought by enhancing chloroplast biogenesis. Similar results have been found by earlier research (Al-Hasany et al., 2019; Spada et al, 2024; Spinelli et al 2010 ; Blunden et al., 1997; Fan et al., 2013; Jannin et al., 2013; Mancuso et al., 2006; Sivasankari et al., 2006).

5.1.1 Water Use Efficiency and Drought Mitigation

Under normal watering conditions, water use efficiency (WUE) was comparable across all treatments. However, during drought stress, significant differences emerged. Among all treatments, SPA+BEM, AMLLK, and Ukelp demonstrated the lowest significant water consumption levels, with recorded values of 762.2 g, 761.0 g, and 760.0 g of irrigated water, respectively. Control plants exhibited a higher demand for water to achieve optimal hydration, whereas bio-effector-treated plants demonstrated reduced water consumption. This indicates improved water retention and drought tolerance mechanisms in the bioeffector-treated plants. These results align with Vafa et al. (2024) who reported similar adaptations in plants treated with seaweed extracts combined with other biostimulants. Bio-effectors containing *Chlorophyta* strains, for instance, illustrated the induction of the production of malondialdehyde (MDA), a secondary metabolite associated with drought stress mitigating properties. Reduced stomatal closure and preservation of photosynthetic pigments were also observed in bio-stimulant-treated plants under drought stress.

It was also reported that the seaweed extracts have secondary metabolites like polysaccharides, flavonoids and phenolic compounds which help the plant in defence mechanism against stresses because of their antioxidative character (Connan et al., 2006; Khan et al, 2009; Rioux et al. 2007; Bakhoun et al., 2023). These compounds accumulated during drought stress are called osmolytes and include Proline, sorbitol, and glycine betaine (Blunden et al, 1986; Khan et al). They can do osmotic regulation of the stomata cells, protect enzymes, act as antioxidants and scavenge free radicals. This prevents the plant from losing water (Singh et al, 2015).

5.2 Nutrient use efficiency

The effect of bio-effector treatments on the mineral composition of durum wheat biomass, expressed in g.kg⁻¹ dry matter, revealed no statistically significant differences ($P \leq 0.05$) in potassium (K), phosphorus (P), calcium (Ca), magnesium (Mg), or zinc (Zn) concentrations among treatments. However, the healthy part showed minimal significant differences for some minerals. In the whole biomass, the bio-effector treatments exhibited some higher potential of some mineral uptake. For instance, SPA+BEM exhibited the highest calcium levels of 3.00 g.kg⁻¹, SPAP4 also was higher in zinc concentration. However, Potassium and phosphorus levels were numerically highest in the control treatment, at 28.61 g.kg⁻¹ and 4.00 g.kg⁻¹, respectively. In the healthy part, the Potassium (K) and phosphorus (P) concentrations varied slightly, with the control showing numerically higher but statistically insignificant values compared to treatments as in the whole biomass. But the rest showed the significance of some bio-effectors. Calcium (Ca) exhibited

significant differences, with AMLLK showing the highest concentration 1.15 g.kg^{-1} and SPAP4 the lowest 0.89 g.kg^{-1} . Magnesium (Mg) ranged from 1.96 g.kg^{-1} (SPAP4) to 2.17 g.kg^{-1} (0.5SPB24P5), both of which were significantly different. Zinc (Zn) concentrations were highest in SPAP4 (0.047 g.kg^{-1}) and lowest in SWP4 (0.032 g.kg^{-1}), with notable statistical differences.

The slight correlation of the findings of our study and the previous studies can be attributed to different factors. Bio-effectors, particularly seaweed-based bio-stimulants, are known to enhance root development, which directly influences water and nutrient absorption (Kapoore et al., 2021). Seaweed extracts are rich in metabolites and trace elements that support plant growth (Stirk and van Staden, 1996; Sivasankari et al., 2006; Khan et al., 2009; Franzoni et al., 2022; Panda et al., 2012). These metabolites, including soluble alginates, protein hydrolysates, and flavonoids, act as chelating agents, facilitating mineral solubility and availability while improving soil aeration and water retention (Khan et al., 2009; Colla et al., 2017; Craigie, 2011). Such mechanisms likely contributed to the observed enhancements in calcium and zinc uptake in the AMLLK and SPAP4 treatments, respectively. The growth-promoting hormones like auxins in bio-stimulants stimulate root elongation and proliferation, promoting nutrient use efficiency (Craigie, 2011; Franzoni et al., 2022). Seaweed extracts also host beneficial microorganisms that engage in symbiotic relationships with plant roots, improving nitrogen fixation and contributing to nutrient supplementation (Rouphael et al., 2017; Mughunth et al., 2024; Deolu-Ajayi et al., 2022; Hurtado et al., 2020).

The study done by EL Boukhari et al. (2020) demonstrated that seaweed-derived bio-stimulants such as *Ascophyllum nodosum*, *Ecklonia maxima*, and *Kappaphycus alvarezii* enhance the uptake of multiple minerals, including K, P, Mg, Mn, and Zn, in various crops. These effects align with findings in this study, where SPAP4 showed a significant increase in zinc content and AMLLK significantly enhanced calcium levels. Such improvements also align with previous reports where *A. nodosum* improved K absorption in wheat (Billiard et al., 2014; Stamatiadis et al., 2015) and *E. maxima* promoted P and Ca uptake in oilseed and tomato crops (Di Stasio et al., 2017; Colla et al., 2017). Seaweed extracts' effects on enhancing mineral uptake are also associated with their ability to mitigate abiotic stresses. They produce secondary metabolites that bind to essential nutrients, enabling their sustained availability to plants under suboptimal conditions (EL Boukhari et al., 2020). Studies have shown that biostimulants containing *Kappaphycus alvarezii* enhanced N, P, K, and S uptake in wheat and maize (Basavaraja et al., 2018; Rathore et al., 2009), which aligns with the increased magnesium content observed in the SPB24P5 of 0.5 ml. l^{-1} treatment in this study.

In addition, Rathore et al. (2008) demonstrated that significant increases in nitrogen (N), phosphorus (P), and potassium (K) uptake were observed only at seaweed extract concentrations of

7.5% or higher. Similarly, Crouch et al. (1990) reported that the application of seaweed extract concentrates on lettuce, under recommended mineral fertilization conditions, significantly enhanced magnesium (Mg), calcium (Ca), and potassium (K) content in the leaves. Furthermore, these findings suggest that seaweed extract application activates genes involved in nutrient transfer from the rhizosphere into the plant. Supporting this, a study by Janin et al. revealed that the application of seaweed extracts containing *Ascophyllum nodosum* induced the overexpression of genes responsible for nitrogen (N) and sulfur (S) uptake. This gene activation, coupled with enhanced root growth facilitated by biostimulants, enables plants to absorb considerable amounts of mineral nutrients when adequate soil nutrient supply is present. Beyond nutrient composition, seaweed-based biostimulants have demonstrated broader growth-promoting effects, including increased biomass and improved vegetative growth in multiple crops. For instance, *E. maxima*-based products enhanced mustard plant yields (Di Stasio et al., 2018), while *A. nodosum* application improved olive yield and oil quality when combined with boron supplementation (Chouliars et al., 2009; Nanda et al., 2022). Such benefits are consistent with the results from Franzoni et al. (2022), who documented the positive effects of kelp extracts on biomass accumulation in willow plants.

6. Conclusion

In summary, while bio effectors positively affected chlorophyll content, water use efficiency, and root biomass, their impact on plant height, nutrient acquisition, biomass production and yield remained limited. Again, despite the observed trends in mineral content, the little statistical significance in the healthy part and lack of statistical difference in the whole biomass indicate that bio-effectors had no distinct effect on the accumulation of macro- and microelements in durum wheat under the conditions of this study. These findings highlight the need for further research to optimize bio effector application strategies, including concentration, strain combinations, and soil conditions, to harness their potential benefits fully. Because several studies confirmed that the application of seaweed extracts bio-effectors is beneficial to both nutrient use efficiency and drought tolerance when there is an adequate supply of favorable conditions and fertilization (Khan et al., 2009) and bio-stimulants can be a cornerstone in the reduction of synthetic chemicals usage in agriculture (Azmi et al., 2023), as EU Farm-to-Fork want to achieve. Our study also proved that the innovative bio-stimulants that are being prepared by climate solutions have the characteristics of influencing drought resilience, some growth parameters, and nutrient use efficiency. However, continuous research must be repeated many times to confirm the availability of these effects on the plant and the soil. After this research, it is worth

conducting further research on these innovative products to quantify the right application dose, their specific effect and the specific plant to be applied.

The program aims to reduce the amount of chemical fertilizers and pesticides without compromising soil fertility in Europe. the bio-effectors can be the best option since they can maintain the soil fertility but also increase productivity. it can also replace or help to reduce synthetic fertilizers and pesticides (Bakhoun.,2023; Azmi et al., 2023; Wang et al., 2018)

The use of algae extracts as a bio-fertilizer can significantly affect the future of agriculture in the world. This type of natural fertilizer not only increases soil and crop productivity but also protects the environment. This type of natural fertilizer can be a suitable alternative to chemical fertilizers and guarantee the improvement of quality and agricultural products (Popko et al., 2018; Wang et al., 2018). Therefore, bioeffectors are recommended to help the European Union reduce the amount of chemicals used in agriculture and plant protection.

7. References

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